

1.3 Constraints

Many of the problems that we have to solve have constraints of one type or another. Take the example of the double pendulum. The lower mass has both a motion that is independent of the upper mass, and one that is tied to it. Using the Newtonian approach to mechanics, we treat the constraints as either normal forces, tensions, stresses, ... Based on the fact that the constraints are treated as forces, Newton's second law can be written to take into account that there are two types of forces, the applied forces, gravity, electric, magnetic, contact (*pushing or pulling*), and the constraint forces, those forces required for the object to follow the prescribed path or stay within the prescribed boundaries

$$m\vec{a} = \vec{F}_a + \vec{F}_c \quad (9)$$

where $\vec{F}_a$ are the applied forces and $\vec{F}_c$ are the forces of constraint.

Since analytical mechanics is based on work (energies), there is no energy associated directly with the constraint force. Therefore, the constraints are introduced as relations between the coordinates. Since there are a variety of different types of constraints, we will examine some of the possibilities and how to incorporate them into the Lagrangian or Hamiltonian. **The simplest constraints are those that can be put into the following form**

$$g_j(q_i) = 0 \quad (10)$$

where g_j can be a function of all the generalized coordinates. Constraints of this form are called **holonomic**. Holonomic constraints can be either time independent (scleronomic) or time dependent (rheonomic). A few examples of Holonomic constraints are

Block sliding on a wedge	$y - ax - b = 0$
Block sliding on an incline plane angle varies	$y - x \tan(\omega t) = 0$
Mass attached to a rod (pendulum)	$r - \ell = 0$
Bead on a rigid wire	$f(q_i) = 0$

All constraints that are not given by $g_j(q_i) = 0$ are called non-holonomic. One example is a particle confined to a box, the constraint is given by $f(\vec{r}) > 0$. Another far more useful example is one that is given in terms of derivatives

$$\sum_i a_{ji}(q_i) dq_i + a_{jt} dt = 0 \quad (11)$$

where the sum is over the total number of generalized coordinates. There are two cases that we have to consider for this type of constraint. The first is if the sum can be written as a total differential

$$\sum_i a_i(q_i) dq_i \quad \text{where} \quad a_i = \frac{\partial g}{\partial q_i} \quad (12)$$

where g is the constraint. This equation can be integrated and therefore put into holonomic form. Note that the condition of integrability is given by

$$\frac{\partial a_j}{\partial q_i} = \frac{\partial^2 g}{\partial q_i \partial q_j} = \frac{\partial a_i}{\partial q_j} \quad (13)$$

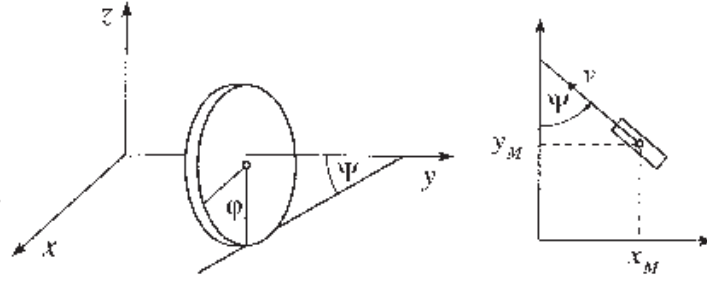


Figure 2: Wheel constrained to roll on a plane surface.

if this condition is not satisfied, then the constraint is non-holonomic. As an example, consider a wheel rolling on a plane (see Fig. 2). The condition for rolling is given by

$$v = a\dot{\phi} \quad (14)$$

where a is the radius of the wheel. The x and y components of the center of mass velocity are given by

$$\dot{x}_m = -v \sin \psi \quad \dot{y}_m = v \cos \psi \quad (15)$$

Combining these equations together, we arrive at

$$\begin{aligned} dx_m + a \sin(\psi) d\phi &= 0 \\ dy_m - a \cos(\psi) d\phi &= 0 \end{aligned} \quad (16)$$

if ψ is a constant, then this equation is integrable, if it is only known after we solve the equations of motion, then this is a non-holonomic constraint.